🚀 **Project ATHENA — Day 7**

Theme: Your First Trading Strategy — Moving Average Crossovers

"A strategy is not an opinion. It's a hypothesis waiting to be tested."

Objective: Learn how to transform an indicator into a complete, testable trading strategy and prepare ATHENA's first backtesting module.

Time Required: ~40 minutes

Today's Big Idea

Yesterday, we built our first indicator:

Moving Average

Today we're going to answer an important question:

When should we actually buy?

This is where most retail traders go wrong.

They think:

Moving Average

↓

Buy

That's incomplete.

A quant thinks:

Indicator

↓

Trading Rule

↓

Backtest

↓

Performance Analysis

↓

Deploy (Maybe)

Notice the last word:

Maybe.

What Is a Trading Strategy?

A strategy is simply a **set of rules**.

Not predictions.

Not intuition.

Rules.

Example:

IF

20-day SMA

crosses above

50-day SMA

THEN

Buy

That's it.

No emotions.

A Complete Strategy Has Five Parts

Every strategy in ATHENA must define these components.

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1. Entry Rule

Example:

20 SMA crosses above 50 SMA

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2. Exit Rule

Example:

20 SMA crosses below 50 SMA

---

3. Position Size

Example:

Risk

=

1%

of portfolio

We've already built the foundation for this with the risk calculator.

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4. Risk Management

Example:

Maximum portfolio exposure

=

25%

or

Stop loss

=

2 × ATR

We'll study ATR later, but the key idea is that risk rules belong to the strategy, not as an afterthought.

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5. Evaluation Metrics

Instead of asking:

"Did it make money?"

We'll ask:

- CAGR
- Maximum Drawdown
- Sharpe Ratio
- Win Rate
- Profit Factor
- Number of Trades
- Average Holding Period

These metrics tell us whether a strategy is robust, not just lucky.

Why Do Crossovers Sometimes Work?

Imagine driving through fog.

The raw road is difficult to see.

A moving average removes some of the visual noise.

When a shorter-term average rises above a longer-term average, it suggests that recent prices are improving faster than the longer-term trend.

This is called a **trend-following** strategy.

It doesn't try to predict the exact top or bottom.

It aims to capture the middle of sustained moves.

Why Crossovers Also Fail

Here's something many tutorials don't emphasize.

During sideways markets:

↑

↓

↑

↓

↑

Moving averages can cross repeatedly.

You buy.

Sell.

Buy.

Sell.

Each trade loses a little.

This is called **whipsaw**.

A strategy can be excellent in trending markets and struggle when prices move sideways.

The lesson:

Every strategy has an environment where it performs well—and one where it doesn't.

Understanding those environments is as important as understanding the strategy itself.

Quant Research Mindset

Instead of asking:

"Is SMA crossover profitable?"

We'll ask:

Under what conditions does it become profitable?

Examples:

- Strong trends?
- High trading volume?
- Low volatility?
- Specific sectors?
- Indian vs. US markets?
- Different moving average windows?

Those are research questions.

Practical Exercise (≈40 Minutes)

Part 1 — Create ATHENA's First Strategy Module (20 min)

Create:

```
src/strategies/

moving_average_crossover.py
```

Define a strategy class:

```
class MovingAverageCrossoverStrategy:

    def generate_signals(self, prices):
        """
        Returns:

        BUY
```

SELL

HOLD

"""

Don't worry about optimization yet.

Keep the interface clean.

Part 2 — Create Sample Data (10 min)

Use the synthetic prices from yesterday.

Calculate:

- SMA(20)
- SMA(50)

Even if you don't yet have 50 observations, think about how your code should behave. Good software handles incomplete data gracefully.

Generate a column:

Signal

Example:

BUY

HOLD

HOLD

SELL

Part 3 — Think Like a Researcher (10 min)

In your journal, answer:

Why might a moving average crossover work better on an index ETF than on a highly speculative small-cap stock?

Think about:

- Noise
- Liquidity
- Trend persistence
- Transaction costs

There isn't a single correct answer, but the reasoning matters.

Architecture Update

ATHENA now has its first end-to-end flow:

Market Data

↓

Indicators

↓

Strategy

↓

Signals

↓

Risk Manager

↓

Portfolio

Next week, we'll close the loop by adding **backtesting**.

Engineering Principle of the Day

Never hardcode strategy parameters.

Instead of:

SMA = 20

Prefer:

short_window

long_window

Later, you'll be able to compare:

- 10/30
- 20/50
- 50/200

without changing the strategy code.

Parameterization is essential for systematic research.

Review Question

Suppose a stock has been moving sideways for three months.

What would you expect from a simple moving average crossover strategy?

A. It will consistently outperform because moving averages always identify trends.

B. It may generate several false buy and sell signals due to frequent crossings.

C. It will stop generating signals entirely.

Answer: B.

This phenomenon is called **whipsaw**, and it's one of the key weaknesses of trend-following systems.